



TM482

Hydraulic servo pump drives

Prerequisites and requirements

Training modules	TM210 – Working with Automation Studio TM400 – Introduction to motion control TM480 – The basics of hydraulics
Software	AS 4.1 or higher
Hardware	---

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1 Introduction

The TM482 training module deals with controlling hydraulic cylinders using targeted movements of hydraulic pumps. The pump movements are controlled using an ACOPOS servo drive and a servo motor. This training will introduce a solution based on the MTHydPump library. This approach will be implemented directly in Automation Studio and tested using a simulated machine.

1.1 Learning objectives

This training module uses selected examples to help you learn about how hydraulic servo pumps work and the steps involved in their commissioning.

- Participants will learn how a servo pump works and about the components of a hydraulic system controlled by a servo pump.
- Participants will learn the parameters needed to size a servo pump drive and understand the underlying control concept.
- Participants will learn how to control a servo pump, calculate energy consumption and protect the components from overload.
- Participants will learn how to configure a servo pump control solution and put it into operation.

2 Structure

One of the most important tasks of hybrid drive systems is to move hydraulic axes. The energy source is a pump that is connected to a variable speed motor. Hydraulic actuators (directional valves) are used to direct the hydraulic fluid to the necessary axis. Hydraulic axes can be either linear cylinders or hydraulic motors. Another important element in these systems is the pressure relief valve, which prevents excessive pressure build-up and thus protects against mechanical damage. System pressure (ActPumpPressure) and motor speed (ActPumpSpeed) are needed as measured variables.

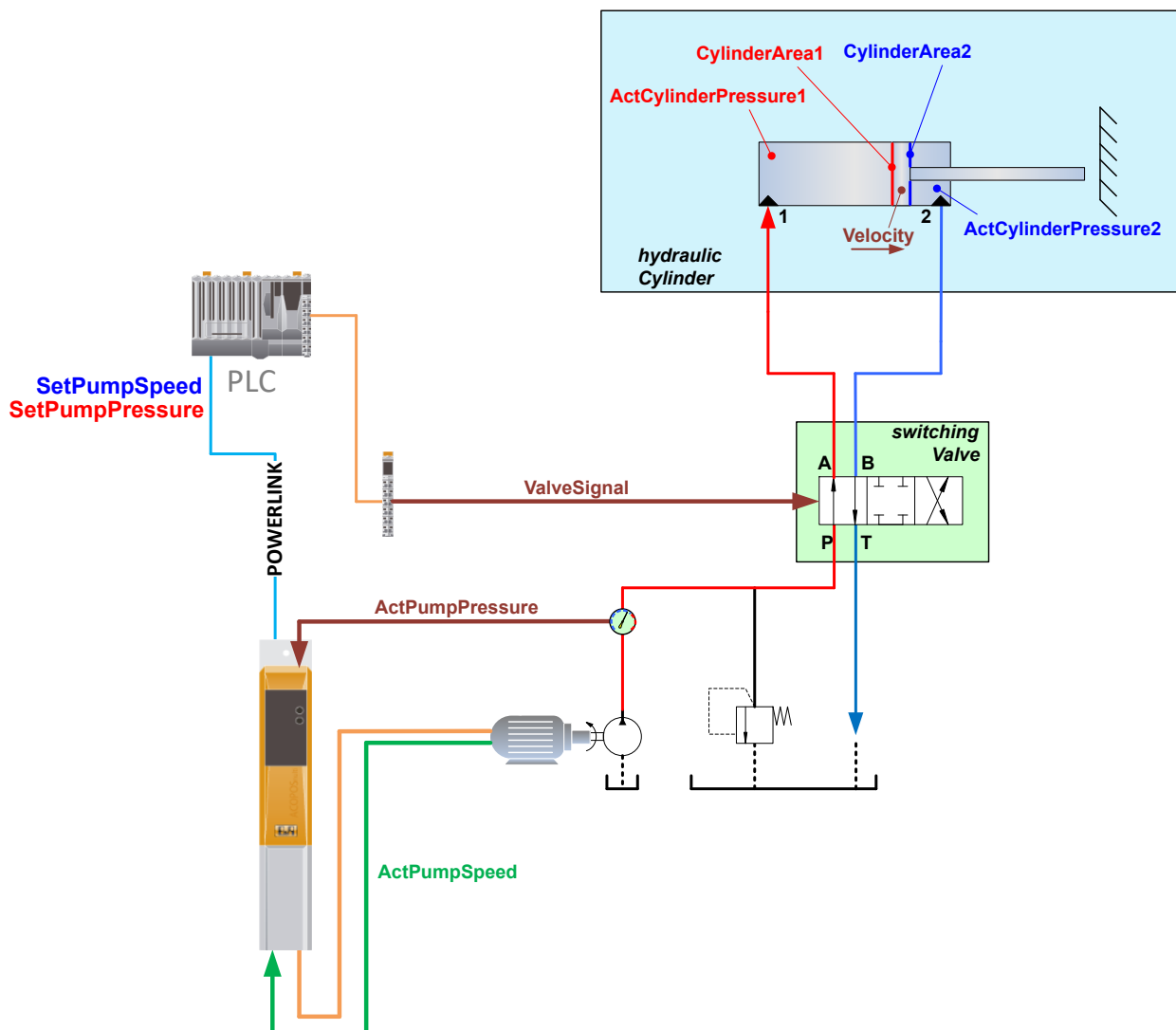


Figure 1: Hardware concept for a hydraulic servo pump drive with an X20 controller, ACOPOS drive, motor and servo pump

3 Components

Hydraulic servo pumps consist of the following three elements:

- Servo motor and hydraulic pump
- Servo drive (ACOPOS)
- Software (MTHydPump)

3.1 Servo motor and pump

This unit is responsible for converting electrical energy into hydraulic energy. The internal gear pump is attached to a synchronous motor via a shaft coupling and housing. There is also a hydraulic pump block with a pressure relief valve and pressure sensor mounted directly on the pump. This equipment is attached to the machine using a mounting flange.



If a pump block is not installed, the pressure sensor should be mounted as close as possible to the pump. In addition, it is essential that a pressure relief valve be included in the system.

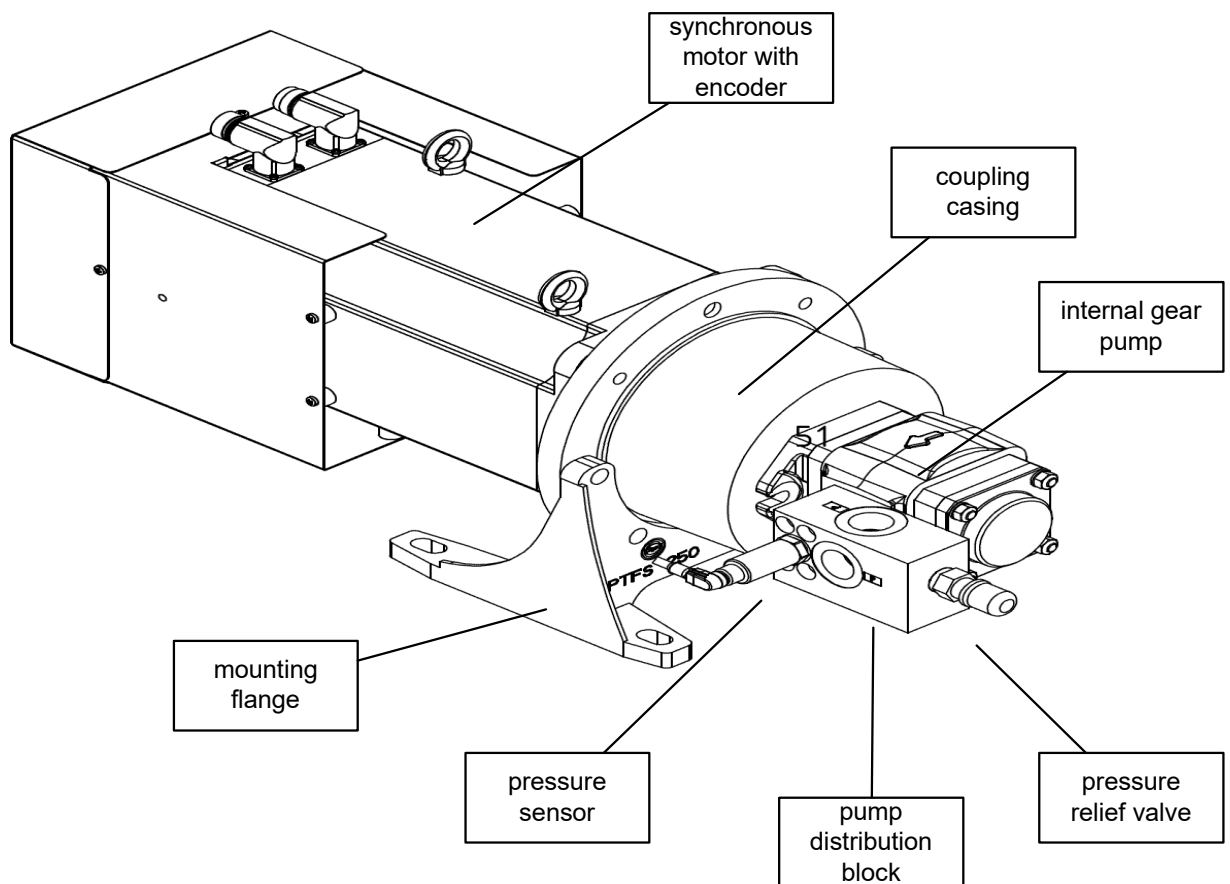


Figure 2: Example of a servo motor / hydraulic pump combination

3.2 ACOPOS



Servo drives at B&R are referred to by the names ACOPOS and ACOPOSmulti. This document will treat ACOPOS as a synonym for B&R servo drives.

ACOPOS drives are used to control the servo motors and therefore the connected pumps as well. This is done by alternately controlling the speed of the servo motor (and the pump) or the pressure. The speed of the motor is read using an encoder card (AC12x in the following image). Pressure is read using an analog input on an AC13x plug-in card.

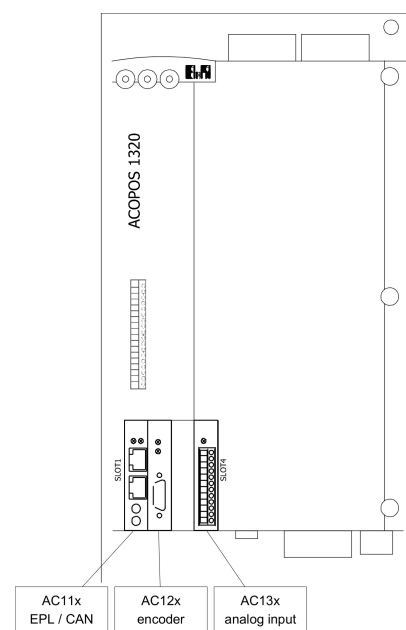


Figure 3: Plug-in cards required on the ACOPOS drive

Since the actual control loop runs on the ACOPOS drive, a very high degree of control quality can be achieved regardless of the performance of the CPU.

3.3 Software

Automation Studio is used to develop the software. The MTHydPumpController function block is primarily used as the main interface and for operation since the control loop itself is running on the ACOPOS drive for performance reasons.



The MTHydPump library is available as an upgrade in Automation Studio 4.1 and is an integral part of Automation Studio in V4.2 and later. This library requires a license and utilizes B&R Technology Guarding technology.

MTHydPump and the ACP10 ACOPOS firmware are dependent on one another. A version of the MTHydPump library is only compatible with a certain ACP10 version. There is a new MTHydPump version for every new ACP10 version.

4 Sizing

An important aspect of the hydraulic servo pump is the correct sizing of the equipment. It is not necessary to be familiar with all equations and relationships between the servo motor, servo drive and pump. It is, however, important to be able to estimate whether the requirements can be met. There are easy standard formulas for this estimate as well as a tool for accurate sizing under consideration of all of the parameters in play.

ID	Unit	Description
n	rpm	Speed
Q	l/min	Flow rate
V _s	cm ³	Displacement volume
η _{vol}		Volumetric efficiency
η _{hm}		Hydromechanical efficiency
M	Nm	Torque
p	bar	Pressure
I	A	Current
k _t	Nm/A	Torque constant

Table 1: Sizing parameters

To determine the required motor speed for a desired volumetric flow rate, it is necessary to know the volumetric flow rate, the pump size and the volumetric efficiency of the pump (where applicable). In addition, the maximum and average motor speed are critical values for motor sizing.

$$n \approx \frac{Q}{1000 \cdot V_s \cdot \eta_{vol}}$$

Another important value for sizing is the torque required to generate a certain pressure. The torque can be easily determined using the relationship between displacement volume and pressure.

$$M \approx \frac{p \cdot V_s}{20 \cdot \pi \cdot \eta_{hm}}$$

The current required at various operating points is also critical when sizing the ACOPOS drive. It can be determined using the motor's torque constant.

$$I \approx \frac{M}{k_t}$$

These three relationships can be used to quickly determine if certain operating states are possible with the given hardware. More precise sizing is possible using B&R's "Servo Pump Sizing Tool" for servo hydraulic systems.

4.1 Servo Pump Sizing Tool

B&R's Servo Pump Sizing Tool is used to size pumps, motors and drives. This tool is specially tailored to applications with hydraulic servo pump drives.



The Servo Pump Sizing Tool is included on the Automation Studio DVD starting with V4.2.2.

4.1.1 Sizing with the Servo Pump Sizing Tool

Sizing is done using a defined machine cycle with the necessary pressure and volumetric flow rate.

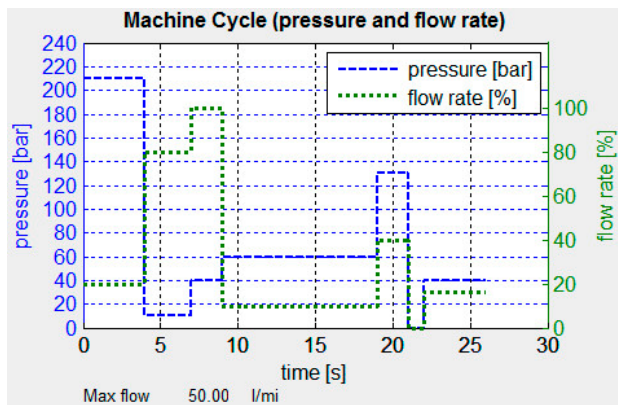


Figure 4: Example of a machine cycle with pressure and volumetric flow rate.

4.1.1.1 Components to be sized

Internal gear pump

- Default: Pumps are available from a hydraulic partner.
- Other pumps can simply be added to the .csv file.

Synchronous motor

- Default: All relevant B&R motors are available with the usual configurations.
- Other motors can simply be added to the .csv file.

ACOPOS servo drives

- Default: All B&R ACOPOS and ACOPOSmulti drives are available.

4.1.1.2 Sizing procedure

- 1) Define a name for the sizing configuration and enter the mains voltage, mains frequency and ambient temperature.
- 2) Enter the machine cycle data including the necessary pressure and volumetric flow rate directly or import it as a .csv file.
- 3) Select the pump
- 4) Select the motor
- 5) Select the drive
- 6) Select Start Evaluation

Figure 5: Servo Pump Sizing Tool

Result

After the evaluation has been started, a concise report is shown with a general overview of the sizing configuration. It is clear here at first glance whether or not the sizing meets the respective requirements.

Report	
Selected pump:	DHPH2-016RK03-1X
Selected motor:	8LSC57.R0022D000-2
Selected ACOPOS:	8V1180.00-2 (passive)
Required torque:	OK
Temperature:	too high
Current ACOPOS:	check Overload

Figure 6: Sizing results report

Clicking on Show Results displays more detailed information. Sizing information is provided for every portion of the machine cycle. In addition, it is also possible to click on Show Diagrams to view a graph that traces motor temperature and the current values.

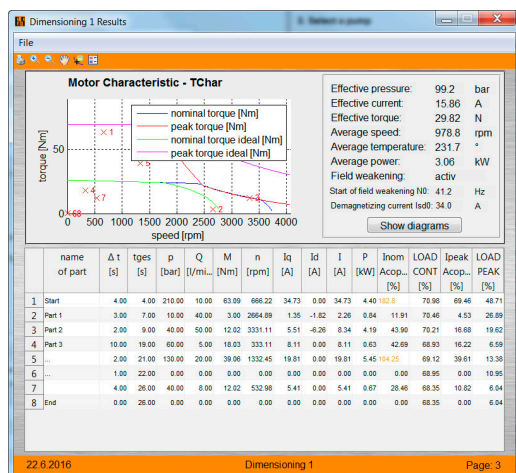


Figure 7: Sizing results for the individual portions of the machine cycle.

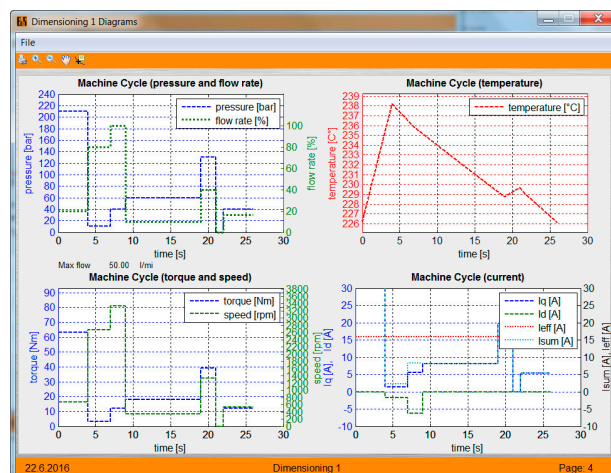


Figure 8: Sizing diagrams

4.1.2 Additional options

- Operation with 2 pumps, 1 motor and 1 ACOPOS servo drive

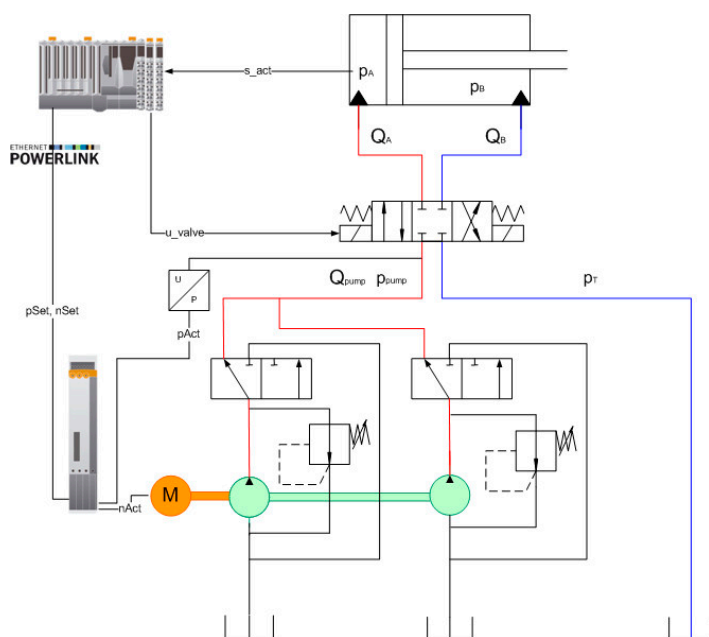


Figure 9: Hydraulic system with 2 pumps, 1 motor and 1 ACOPOS drive

- Field weakening
- Machine cycle defined via the torque and speed.
- Sizing configuration without a pump. Sizing configuration for applications with only a motor and ACOPOS drive.
- Creating a PDF report that documents the entire sizing configuration.

4.1.3 Exercise 1 - Servo Pump Sizing Tool

In this exercise, the Servo Pump Sizing Tool will be looked at in more detail.

- a) Open the Servo Pump Sizing Tool
- b) Import the machine cycle example (example_export_import_cycle.csv)
- c) Complete sizing operation
- d) Create PDF report
- e) Analyze and discuss the various results

4.1.4 Exercise 2 - Servo Pump Sizing Tool

In this exercise, the Servo Pump Sizing Tool will be looked at in more detail.

- a) Open the Servo Pump Sizing Tool
- b) Sizing for the following machine cycle is needed.

Machine Cycle				
name of part	time [s]	abs. time [s]	pressure[bar]	flow rate [l/min]
Start	2	2	10	14
Part 1	2	4	300	2
Part 2	1	5	300	0
Part 3	3	8	10	14

Figure 10: Machine cycle example

- c) Complete sizing operation
- d) Create PDF report
- e) Analyze and discuss the various results

5 Software concept

For typical servo pump applications, pressure changes in a matter of milliseconds; it is therefore necessary to run the control loop with a very fast cycle time. It is also important that the values for the current pressure and current speed be made available with the least possible amount of delay. This is why the position controller has been replaced by a pressure controller (PQCTRL) on the ACOPOS drive. With this approach, the control loop directly on the ACOPOS servo drive is calculated using a 400 μ s cycle.

To make things as easy as possible, configuration and setpoint specification are handled exclusively on the controller using the MTHydPumpController function block.

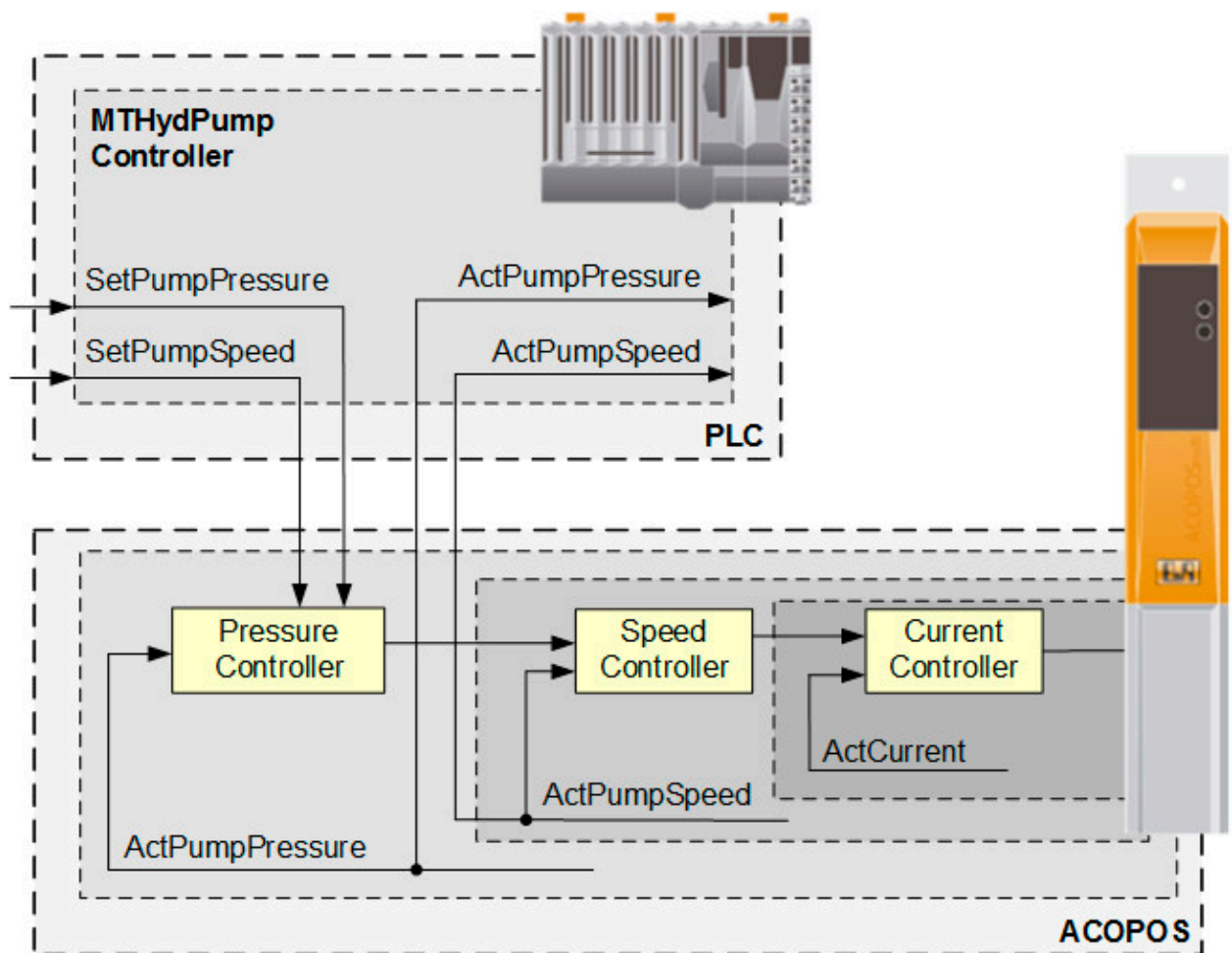


Figure 11: Overall approach to closed-loop servo pump control

POWERLINK and CAN are available as interfaces between the ACOPOS drive and controller. Setpoints and actual values for speed and pressure are written to and received from the ACOPOS drive cyclically. The function block can implement pre-filtering before the setpoints are sent to the ACOPOS drive.

5.1 Closed-loop control concept

The actual control loop for the servo pump is an alternating pressure/speed controller. There are therefore two setpoints – pressure and speed. A setpoint for the flow rate in l/min can be converted directly into a speed setpoint (see "Sizing" section). The existing ACOPOS speed controller is used here. The PQCTRL function block is used as the pres-

sure controller to replace the position controller. A speed is provided on the output of the PQCTRL function block in order to control the pressure setpoint or set the speed setpoint to nSet. The following block diagram shows the control concept for the pressure/speed controller on the ACOPOS drive.

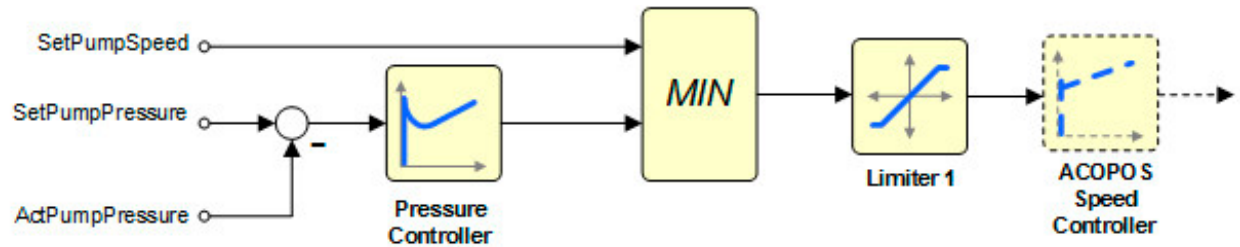


Figure 12: PQCTRL control concept on the ACOPOS drive

6 MTHydPump library function blocks

6.1 General information

The MTHydPump library includes function blocks for controlling and monitoring a servo pump. An ACOPOS drive that is connected to a controller via POWERLINK is required here.



Each version of the MTHydPump library is only compatible with a certain version of the ACP10_MC library.

6.2 MTHydPumpController

The MTHydPumpController function block is used to control servo pumps. Similar to a PLCopen function block, it functions as an interface to the ACOPOS drive.

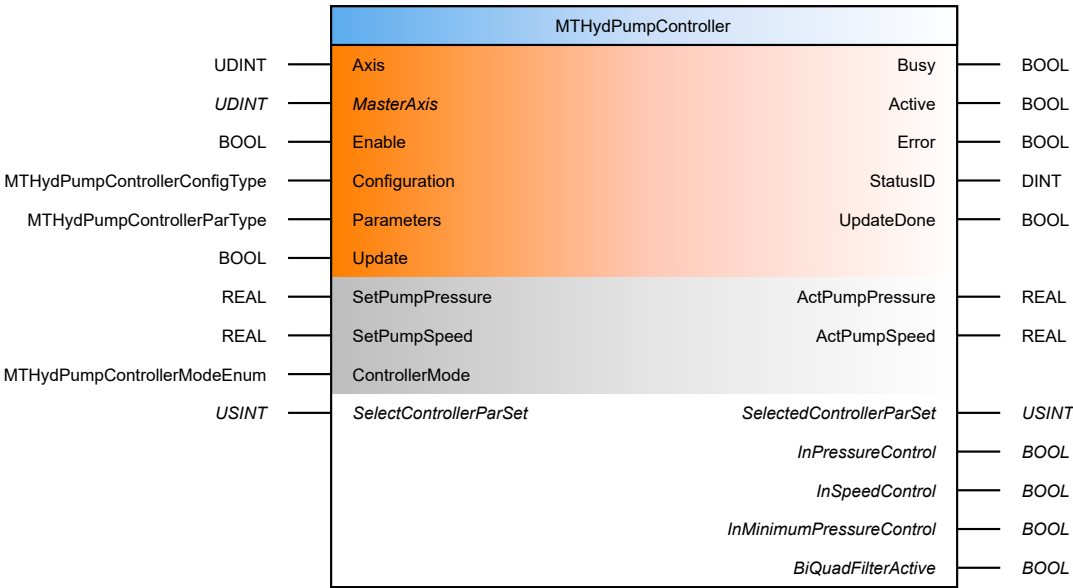


Figure 13: The MTHydPumpController function block



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It is necessary to bring the ACOPOS drive to a standstill (PLCopen state) and switch on the control structure on the ACOPOS drive. This is done using the PLCopen function block MC_Power. Function block MTHydPumpController is only fully functional in the Standstill axis state! After the MTHydPumpController function block has switched off the position controller, the user is not permitted to switch it back on. Axis handling is therefore a prerequisite for the MTHydPumpController function block.

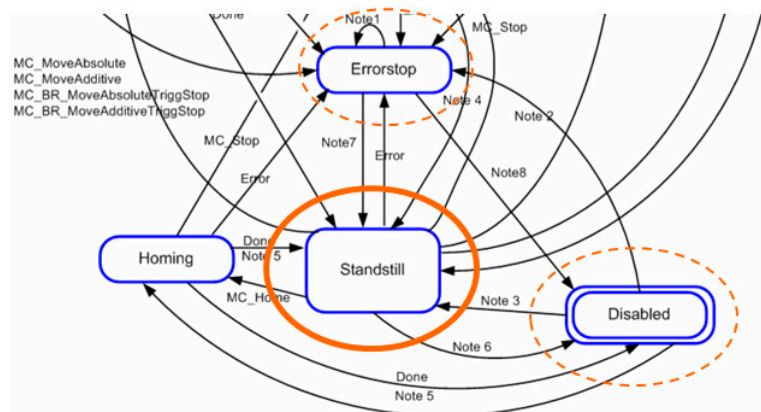


Figure 14: Process from the perspective of PLCopen

The pressure/speed controller on the ACOPOS drive is created and configured once the MTHydPumpController function block is enabled. It then goes into a cyclic state where the setpoints and actual values are transferred cyclically. When changes occur within the parameter structure, a rising edge on the Update input is needed to transfer the values acyclically to the ACOPOS drive.

The system is configured using the two input structures Configuration and Parameters. These structures contain general parameters and limits that can be taken from data sheets as well as parameters that need to be identified during commissioning.

Parameter structure:

- General parameters
- Sensor scaling
- Control loop parameter sets (0-9)

Parameters that need to be determined

- Filter time for the pressure signal Parameters.ActPressureFilterTime
- Controller parameters Parameters.ControllerParameterSet[].PressureController.ProportionalGain, -.IntegrationTime, -.DerivativeTime, -.FilterTime
- Maximum negative speed Parameters.ControllerParameterSet[].MaxReversePumpSpeed
- Acceleration limit Configuration.MaxPumpAcceleration



The parameter Configuration.MaxPumpAcceleration must be determined using the calculation tools or calculated by experts before commissioning the machine. Commissioning without an acceleration ramp can result in cavitation that can lead to the destruction of the pump!

The function block also provides master-slave functionality for pairing multiple axes. These functions are described in more detail in the Additional functionality section.

6.3 MTHydPumpPowerMeter

The MTHydPumpPowerMeter function block does not handle any functions actively. It is only used to display and calculate electrical, hydraulic and mechanical power. The electrical and mechanical power values are read directly from ACOPOS ParIDs. The hydraulic power is calculated using the current pressure, current speed and pump parameters.

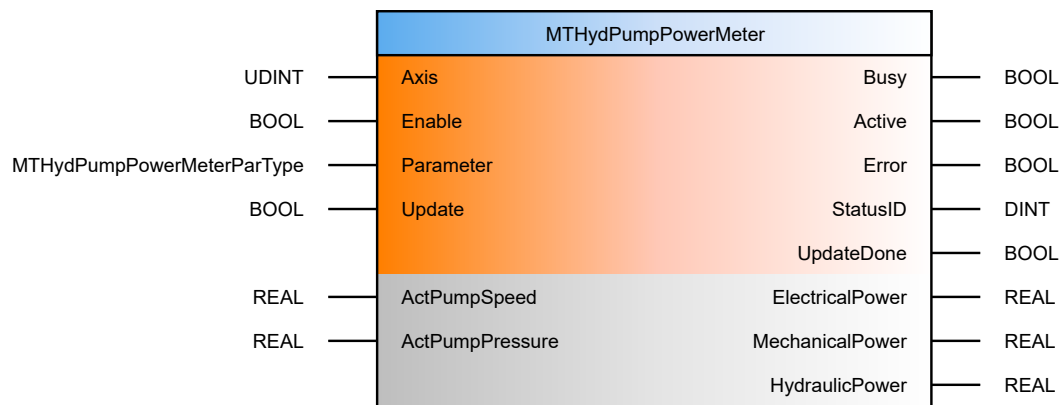


Figure 15: The MTHydPumpPowerMeter function block



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6.4 MTHydPumpDriveProtection

The MTHydPumpDriveProtection function block provides overload protection for the ACOPOS drive, motor and pump. Since a cutoff of the ACOPOS drive (and thus the motor) during operation must be avoided, this function block limits the maximum values for torque and speed to the nominal values that are possible for continuous operation when it appears that the ACOPOS drive and motor may be overloaded. To protect the pump from thermal damage when it is in the overload range, the maximum torque is reduced down to 0 Nm.

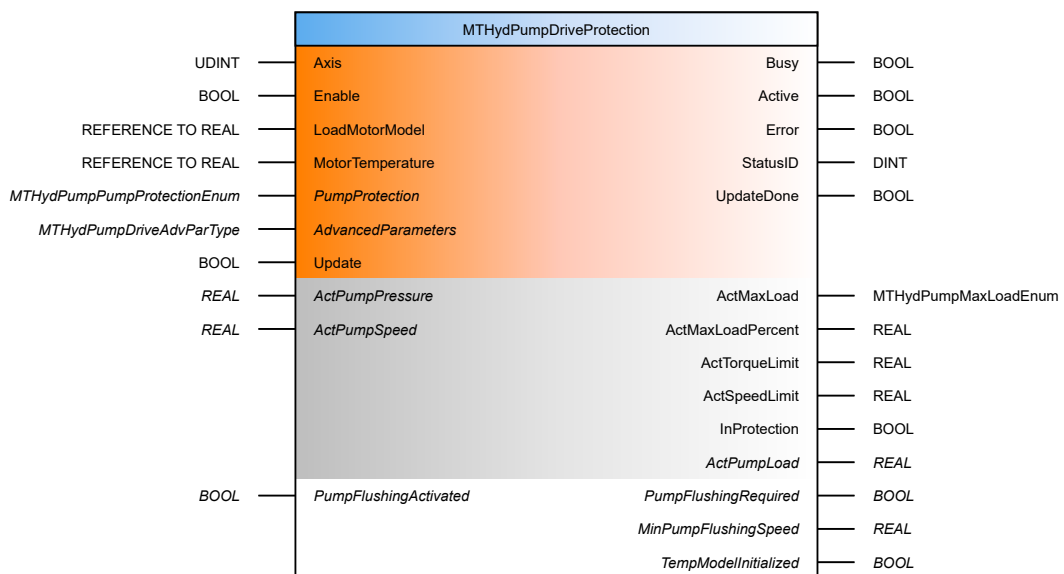


Figure 16: The MTHydPumpDriveProtection function block



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What the MTHydPumpDriveProtection function block does can be divided into two parts. The first part provides overload protection for the ACOPOS drive and motor. The second part provides overload protection for the pump.

6.4.1 ACOPOS drive and motor protection function

Function block MHydPumpDriveProtection checks multiple load parameters on the ACOPOS drive. As soon as a load limit is exceeded (90% by default), the function block automatically limits the torque and speed on the ACOPOS drive.

The following parameters are checked cyclically:

- Continuous current
- Peak current
- Power
- Heat sink temperature
- Motor temperature
- Temperature of the motor model
- Junction temperature (IGBTs)

These default limits for the load parameters can be changed in parameter structure AdvancedParameters.

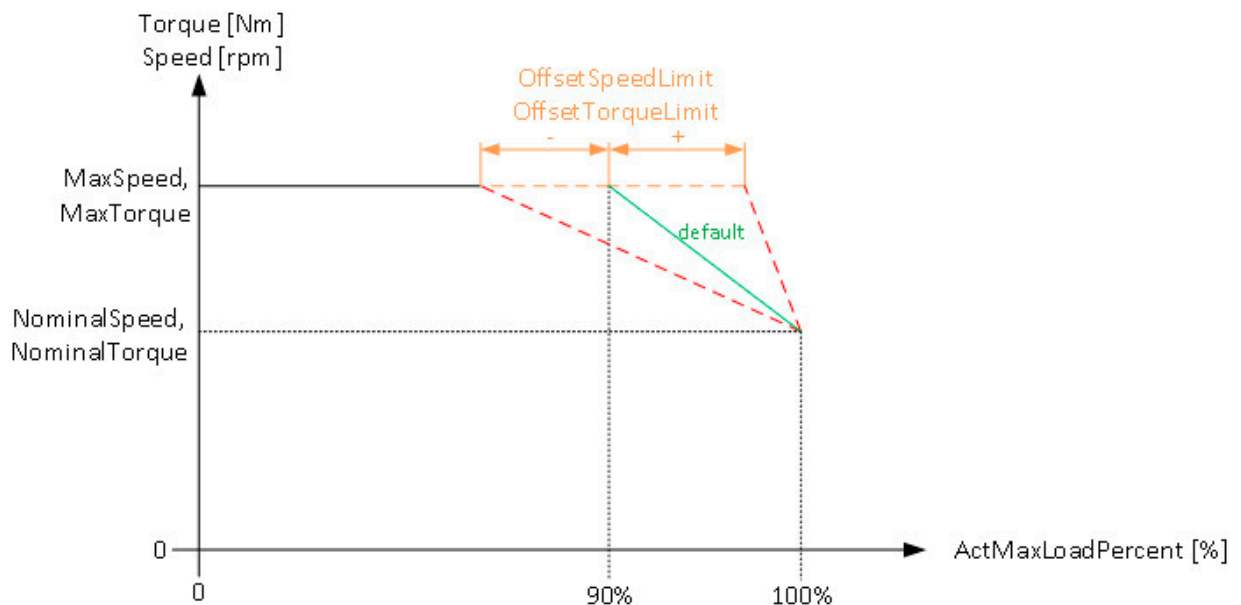


Figure 17: Limiting torque and speed on motors and ACOPOS drives

6.4.2 Pump protection function

To protect the pump, a thermal pump model is executed in the function block that calculates the pump's thermal load.

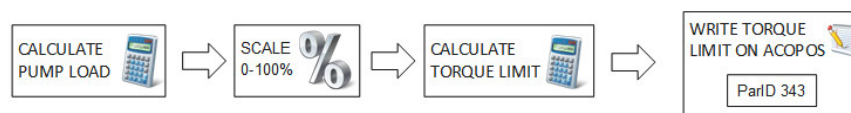


Figure 18: Procedure for pump protection

The thermal load of the pump is calculated in the MHydPumpDriveProtection function block on the PLC and scaled to a range from 0–100%. Torque is reduced if the load exceeds a certain temperature, which brings the motor to a standstill in certain cases. To prevent this worst case scenario, output PumpFlushingRequired signals that the thermal load has exceeded the load limit resulting in a reduction. In addition, output MinPumpFlushingSpeed indicates the optimum speed for flush mode. To induce thermal cooling, it is necessary to flush the pump at a minimum flow rate. The load is reduced by a certain percentage depending on the flush duration or flush volume. If torque is reduced to

0%, the PumpFlushingActivated input must be signaled that cooling measures (flushing) are being taken since this input is used to disable torque limiting.

6.4.3 Initializing the motor temperature model

During operation, the motor temperature model is calculated cyclically on the ACOPOS drive. If the controller is restarted, the values calculated for the motor temperature model are discarded and the ACOPOS drive begins calculations again at 0.0% load.

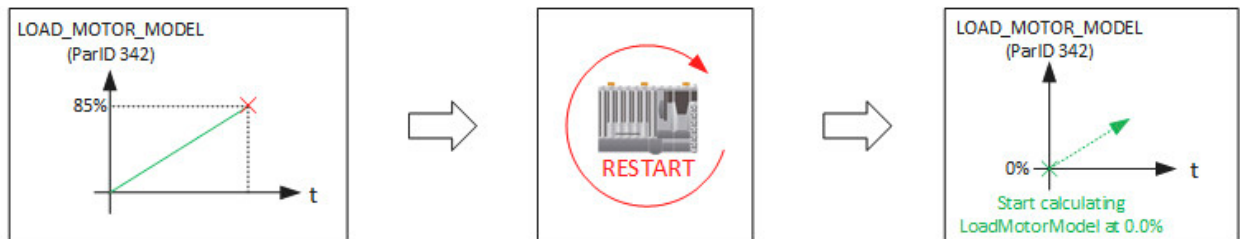


Figure 19: Motor temperature model not applied.

The MTHydValveDriveProtection function block initializes the motor temperature model on the ACOPOS drive with preset values when the controller is restarted. In order to initialize the temperature model, the axis must be in the "Disabled" PLCopen axis state, the LoadMotorModel and MotorTemperature inputs must be connected correctly and their values must be in the valid range. If not, a warning is output and the motor temperature model will not be initialized (TempModelInitialized = FALSE).

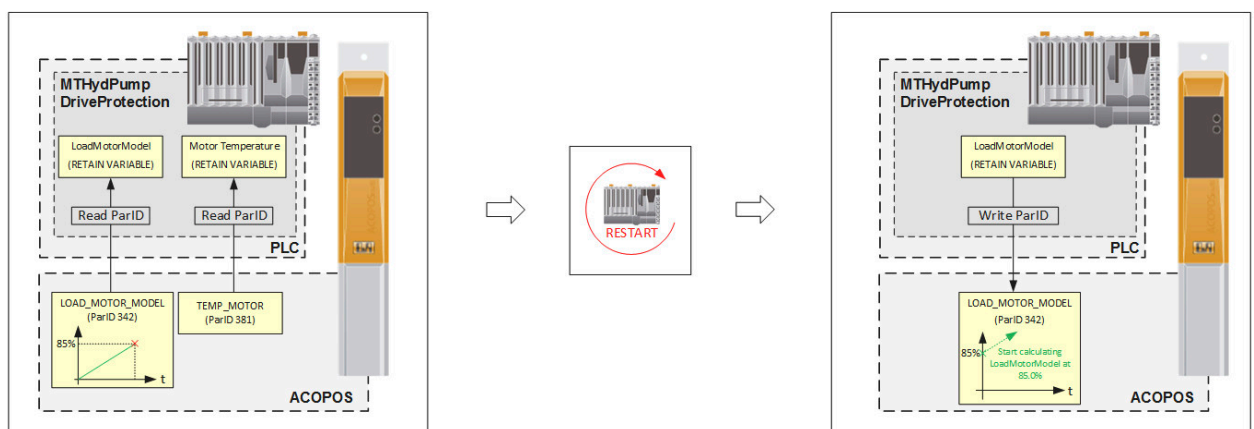


Figure 20: Motor temperature model applied.



The motor temperature model can only be initialized when the axis is in the "Disabled" PLCopen state.

6.5 MTHydPumpSimulationModel

The MTHydPumpSimulationModel function block is used to simulate a hydraulic servo pump. This system comprises a servo pump, a switching valve and a hydraulic cylinder. The function block is already initialized with certain default values so a specific hydraulic system is simulated right away when you call the function block. The individual parameters can be modified as desired allowing any other hydraulic system to be simulated.

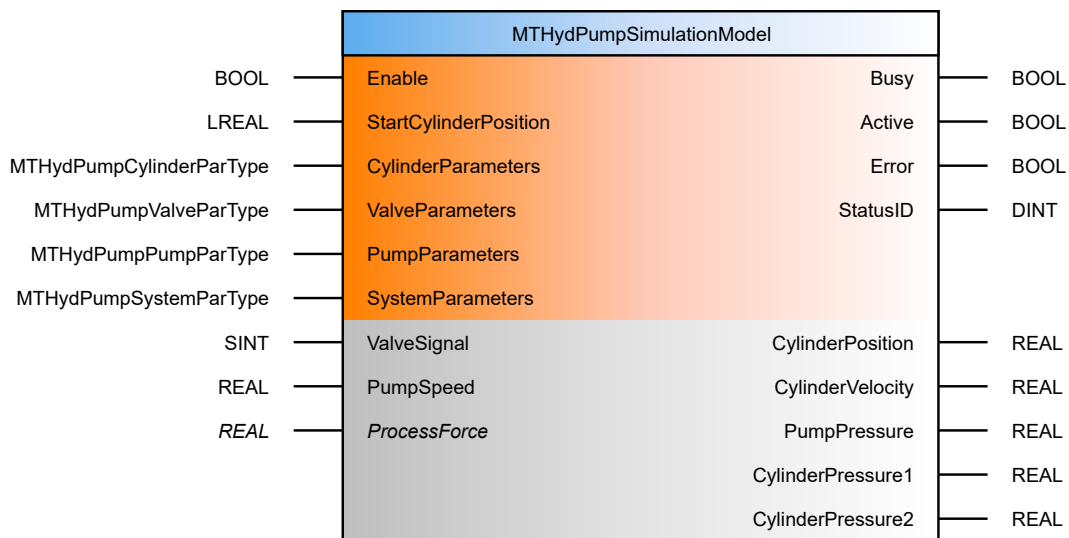


Figure 21: The MTHydPumpSimulationModel function block



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6.6 MTHydPumpPressureTuning

Function block MTHydPumpPressureTuning automatically configures the pressure controller on the ACOPOS drive.

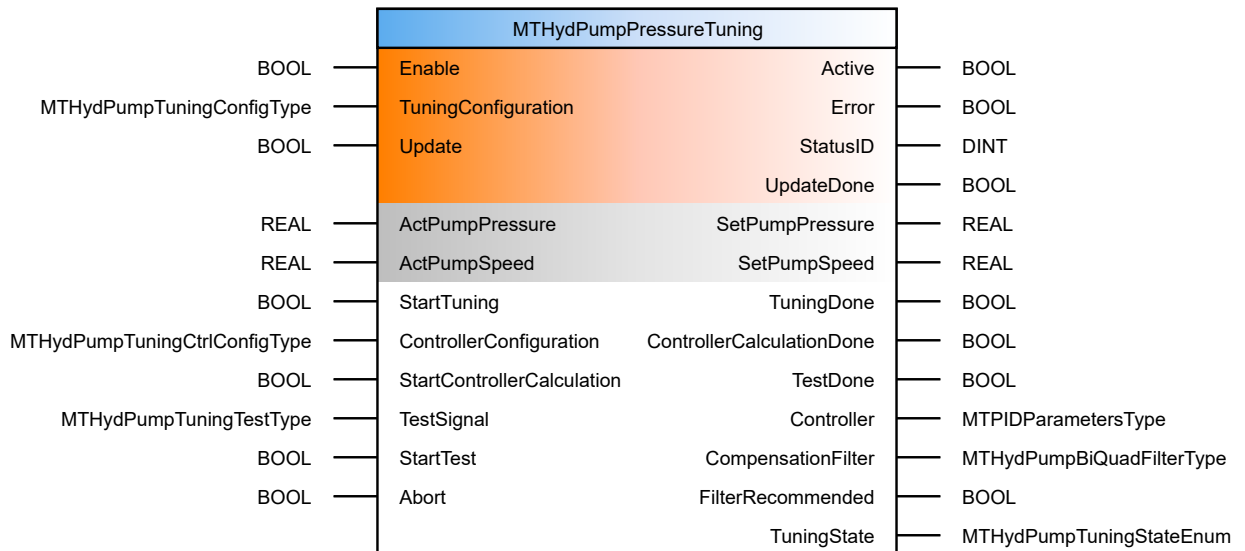


Figure 22: Visual representation of function block MTHydPumpPressureTuning



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Prerequisites for autotuning

To complete tuning, function blocks MTHydPumpPressureTuning and MTHydPumpController must be linked to each other.

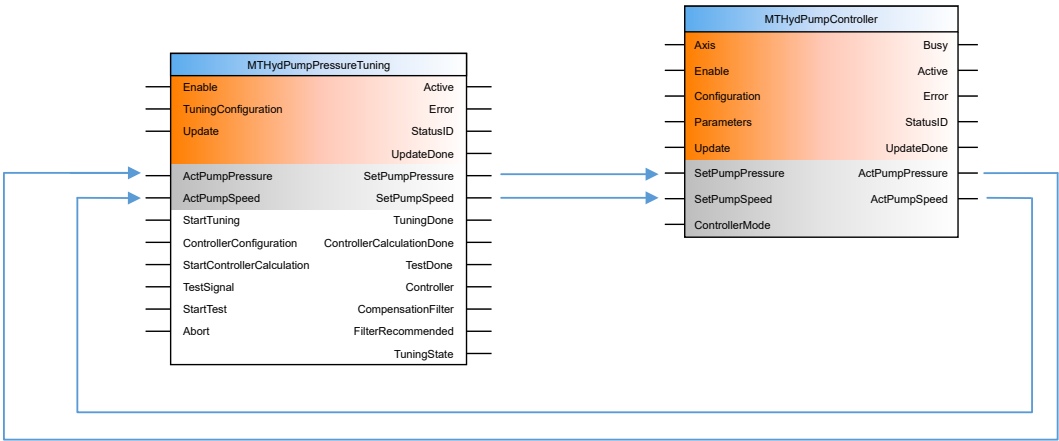


Figure 23: Connecting the function blocks

It is also necessary to ensure that the cylinder is in the position where pressure control should be carried out (cylinder is pressing against the material and the valve is open).

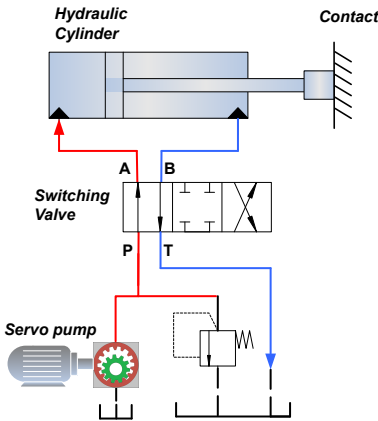


Figure 24: Cylinder in contact position

Tuning process

- The hydraulic system is excited with a certain rotary speed signals (jump signals and harmonic signals). These excitation signals can be configured on the function block.
- The pressure is also measured (system response)

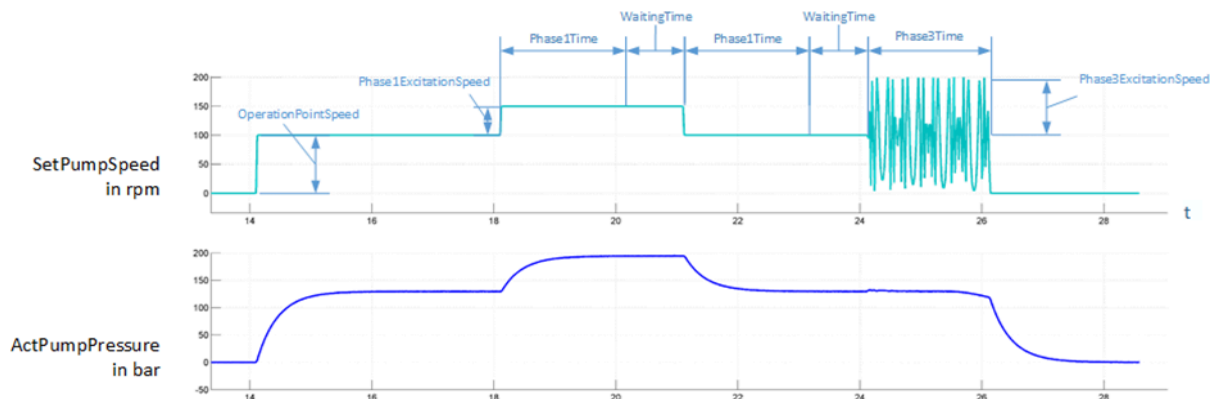


Figure 25: Excitation signal for pressure controller tuning

- With the excitation signal and the measured pressure, the function block identifies the dynamic system and calculates the frequency response (Bode plot). The resonant frequency is also identified.

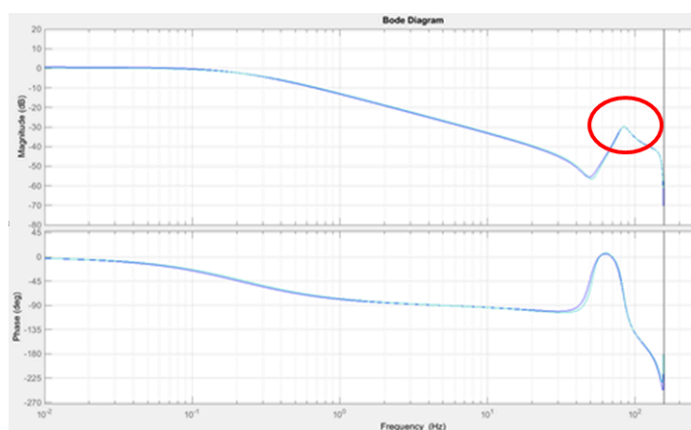


Figure 26: Bode plot

- After the system has been successfully identified, a PI or PID controller and a compensation filter is automatically created.
- Command StartTest can be used to verify the calculated PID parameters based on pressure jumps. The test signal can be configured on function block MTHydPumpPressureTuning.

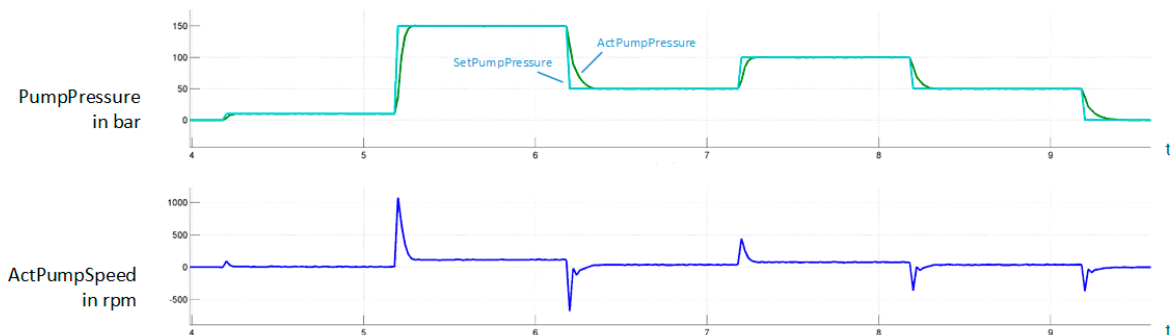


Figure 27: Test signal to check the controller parameters

- If necessary, fine tuning can also be carried out. This is done by changing the controller configuration (RiseTime, ControllerType, DerivativeTimeScale, FilterTimeScale) and starting the new controller parameter calculation with command StartControllerCalculation. The newly calculated parameters can then be verified again using the test signal. The following fine tuning procedure has proven to be successful in real-world situations:
 - Too slow: Reduce → RiseTime
 - Overshoot is too large: Increase → PID controller instead of PI controller, DerivativeTimeScale
 - If FilterRecommended = TRUE → ActivateCompensationFilter = TRUE



It is necessary to ensure that during tuning the current pump pressure (ActPumpPressure) does not exceed the maximum pump pressure (MaxPumpPressure). Tuning is canceled in this case.

6.7 Using function blocks

Keep the following in mind when using the function blocks provided in the MTHydPump library:

- Before you can use the MTHydPumpController function block for speed or pressure control, the axis must be in the PLCopen state "Standstill". Otherwise the function block is enabled, but closed-loop control for setpoint values is not possible because the controller has not yet been enabled on the ACOPOS drive.
- If function block MTHydPumpController is used together with MTHydPumpDriveProtection, the following call sequence must be adhered to in order for both to work properly:

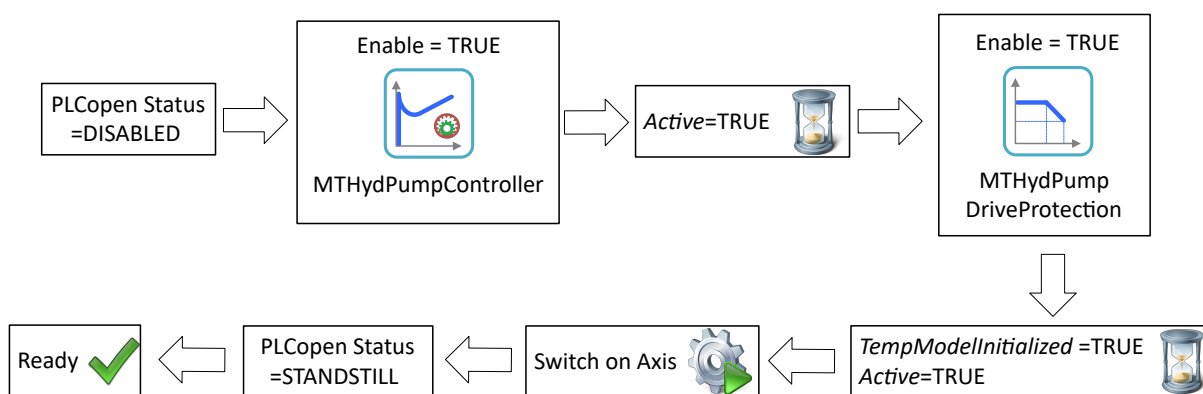


Figure 28: Function block connection with StandaloneMode = mtPUMP_WITH_PRESSURE_CONTROL

- If function block MTHydPumpDriveProtection is used in stand-alone mode, the following call sequence must be adhered to in order for it to work properly:

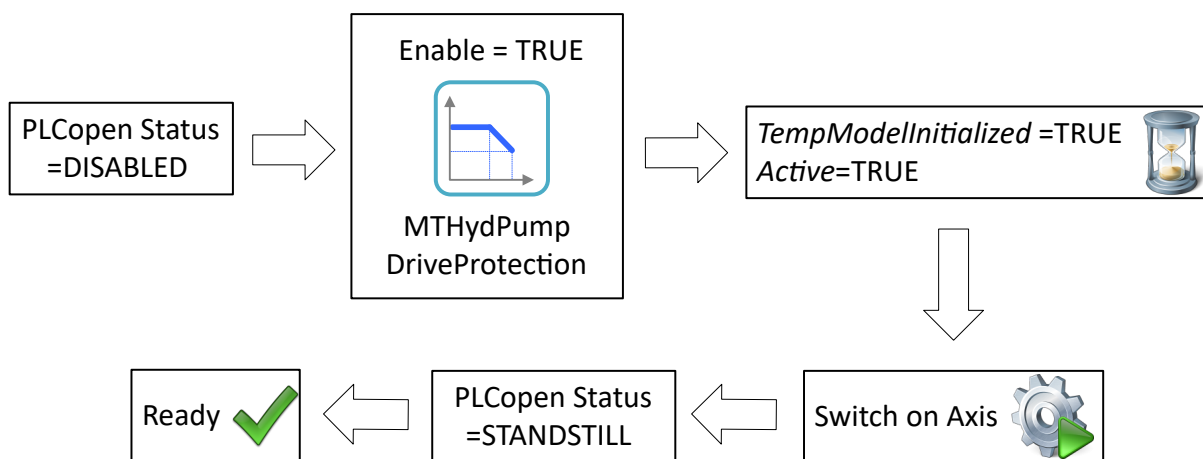


Figure 29: Function block connection with StandaloneMode = mtPUMP_WITHOUT_PRESSURE_CONTROL

7 Technology solution – Hydraulic servo pump control

The B&R mechatronic solutions, and also Technology Solution Hydraulic Servo Pump Control, are fully integrated in Automation Studio. The result is a simple and uniform tool for developing projects as well as commissioning closed-loop control processes and machines.

This Technology Solution provides the user a preprogrammed solution in the form of a package containing all the necessary programs, variables and data type declarations as well as an HMI application and a simulation model. The programs are designed so that they can be added to and executed in any existing project. Users can also create their own template projects or startup projects and manage them in a separate directory.

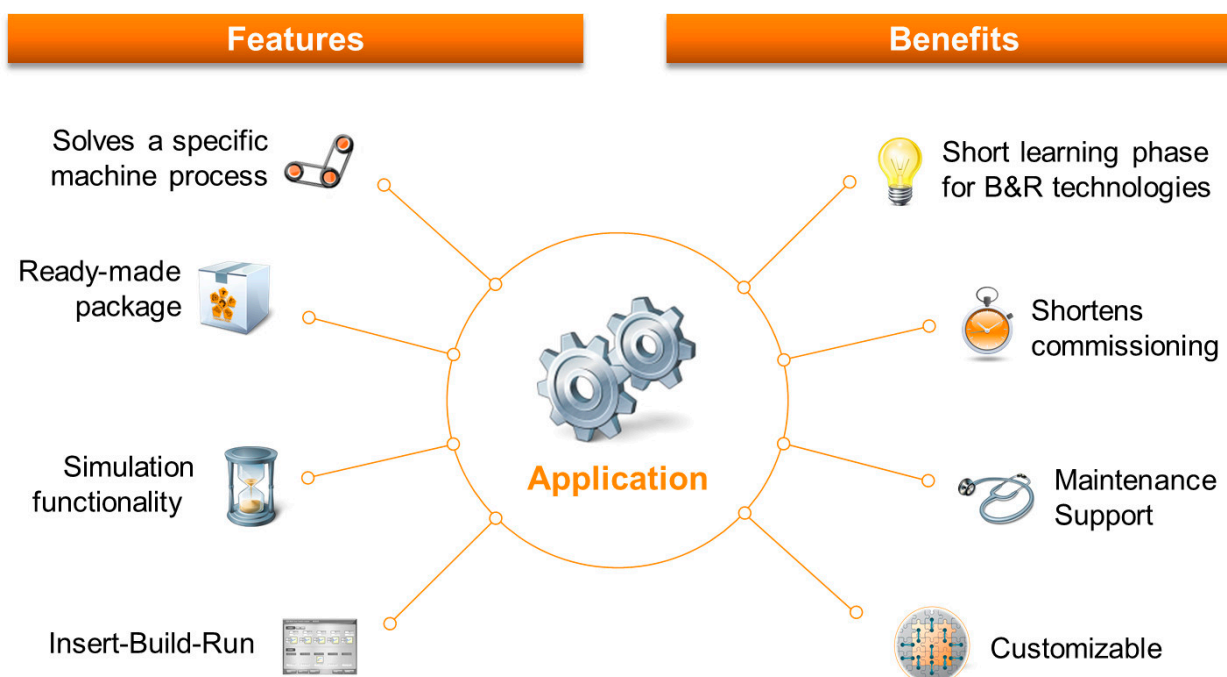


Figure 30: Technology Solution – Description

Technology Solution Hydraulic servo pump control includes speed and pressure control for a hydraulic system with a cylinder, switching valve and servo pump. It is possible to operate the servo pump in two different modes:

- Speed control
- Alternating speed/pressure control

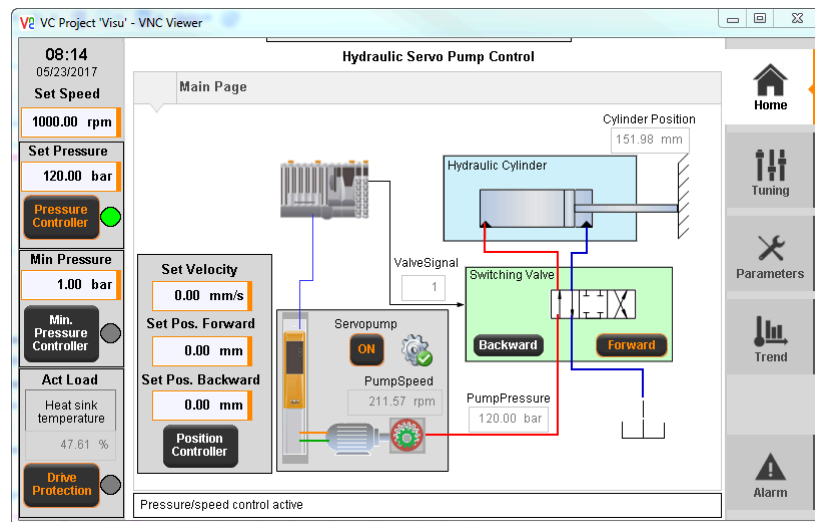


Figure 31: Start page for the Hydraulic Servo Pump Control Technology Solution HMI application.

7.1 Technology Solution exercise

In this exercise, Technology Solution Hydraulic Servo Pump Control is out into operation.

- a) Inserting the Technology Solution in a new project.



Figure 32: Insert the Technology Solution

- b) Activating the HydraulicServoPumpControl configuration

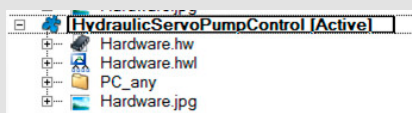


Figure 33: Activate the configuration

- c) Transfer the project to the controller (ARsim)
- d) Connect to the HMI application (VNC viewer, IP: 127.0.0.1).
- e) Execute speed control or alternating pressure/speed control.
- f) Become familiar with implementation of the Technology Solution. How are the function blocks configured, and how is the application linked with the simulation model, etc.?

7.2 ACOPOS drive cutoff protection exercise

In this exercise, the ACOPOS drive cutoff protection provided in function block MTHydPumpDriveProtection will be tested using Technology Solution Hydraulic Servo Pump Control.

- Use the previous project with Technology Solution inserted.
- Using the HMI application (VNC Viewer), move the cylinder into the material and use closed loop control to maintain a pressure of 200 bar.
- Hold pressure until function block MTHydPumpDriveProtection begins to limit maximum torque and speed.
- View Trend to monitor how speed limiting reduces the current pressure and the pressure setpoint can no longer maintained.

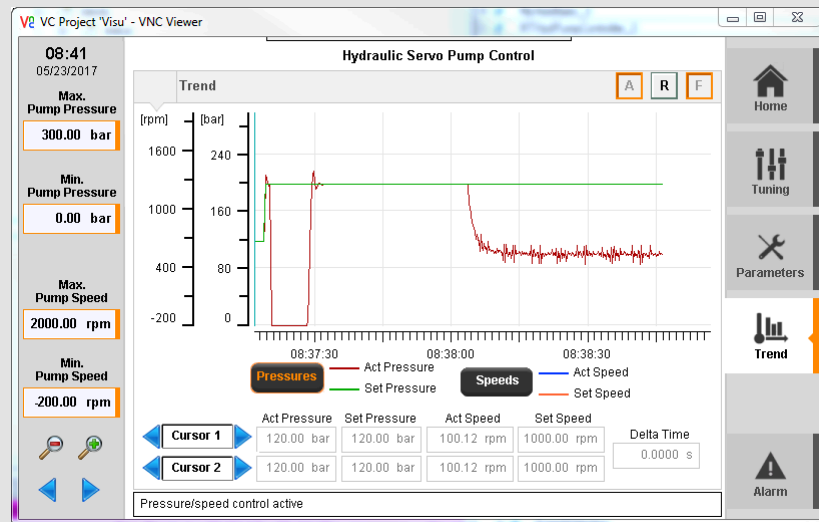


Figure 34: Trend shows speed reduction



Through the simulation, it is possible that the pump speed starts to oscillate during torque limiting.

7.3 Exercise - Speed Torque Chart

In this exercise, a load cycle will be recorded with Technology Solution Hydraulic Servo Pump Control and drive sizing will be checked in the Speed Torque Chart.

- Configure and start motion trace with ParIDs 251 (current speed) and 277 (motor torque).
- Execute multiple movements with the cylinder using the HMI application (VNC Viewer).
- Stop motion trace and save the trace as a csv file.
- Open the Speed Torque Chart in Automation Studio.
- Add the load cycle that was just recorded.

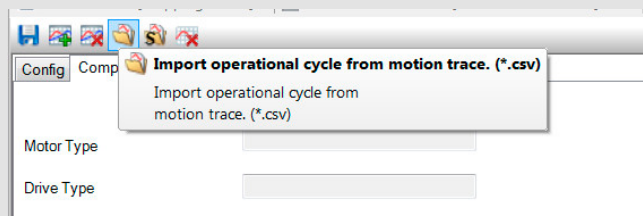


Figure 35: Add the load cycle to the Speed Torque Chart as a .csv file.

- Check the drive sizing using the Speed Torque Chart (operating point, torque reserves, thermal reserves, speed reserves).

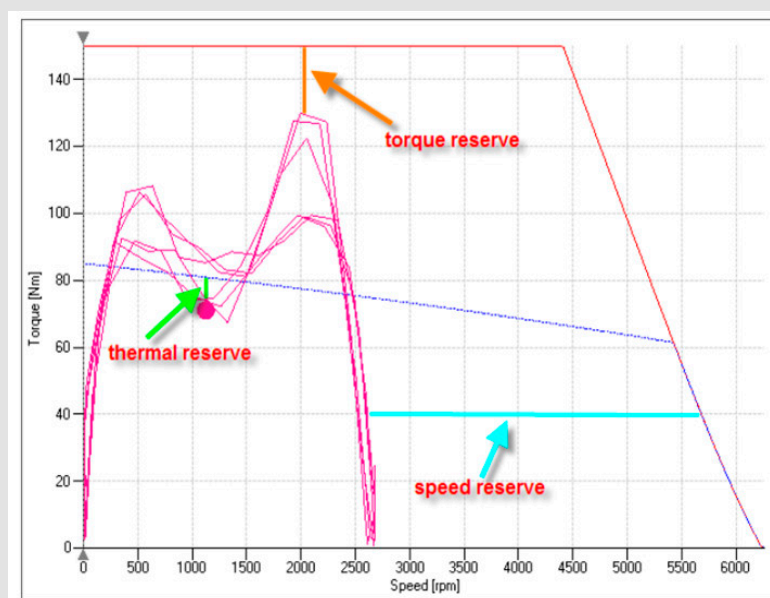


Figure 36: Speed Torque Chart

7.4 Exercise – Power monitoring

In this exercise, the simulated power values in Technology Solution Hydraulic Servo Pump Control is monitored using function block MTHydPumpPowerMeter:

- Use the previous project with Technology Solution inserted.
- Using the HMI application (VNC Viewer), move the cylinder into the material and use closed loop control to maintain a pressure of 120 bar.
- Go to Automation Studio.
- Open "Watch Window" for task HydCtrl.
- Add function block MTHydPumpPowerMeter_0.
- Monitor the electrical, mechanical and hydraulic power.
- Move the cylinder and monitor the change to the power values.

7.5 Exercise – Pressure controller tuning

In this exercise, the pressure controller on the ACOPOS drive is automatically set up using Technology Solution Hydraulic Servo Pump Control.

- a) Use the previous project with Technology Solution inserted.
- b) Using the HMI application (VNC Viewer), move the cylinder into the material and use closed loop control to maintain a pressure of 120 bar.
- c) Leave the valve open.
- d) Execute command `StartTuning` on the Tuning HMI page.
- e) Observe tuning in the Trend report.
- f) Check the calculated controller parameters with the test signal.
 - In Automation Studio, install a Trace with inputs `ActPumpPressure` and `ActPumpSpeed` and outputs `SetPumpPressure` and `SetPumpSpeed` on function block `MTHydPumpPressureTuning`.
 - Start the test signal using `StartTest` on the Tuning HMI page.
 - After the test signal, verify the controller quality in Automation Studio using the Trace function.
- g) If necessary, change the controller configuration (adjust `RiseTime`, `ControllerType`, `DerivativeTimeScale`, `ActivateCompensationFilter`) and recalculate the new parameters.

8 Master/Slave mode

Another important function is master/slave mode. This functionality can be used to create multi-pump systems. To use master/slave mode, the reference to the master must be specified for the slave system function blocks. Cross-communication is used to transfer the master's speed specifications to the slaves as well. The `ControllerMode` input on the function block can be used to switch master/slave mode on and off. This allows the slave axis to be linked and unlinked as required. With an appropriate hydraulic system, both can also work as standalone systems and carry out independent movements at the same time.

Required configuration of the slave pump

- 1) Axis
- 2) MasterAxis
- 3) `ControllerMode = mtPUMP_SLAVE_MODE`
- 4) `Configuration.SlaveGearRatio`
- 5) `Configuration.MaxPumpAcceleration`
- 6) `Configuration.MaxPumpSpeed`
- 7) `Configuration.ReversePumpMotorSpeed`
- 8) `Parameters.ControllerParameterSet[x].MaxReversePumpSpeed`
- 9) `Parameters.DisableMinPressureController = TRUE`



Positive limitation of the speed setpoint **Configuration.MaxPumpSpeed** is **not enabled** in Slave mode!

Speed controller settings – master/slave

Tune the speed controller – Each axis separately!
See section 10.2.2 "Tuning the ACOPOS speed controller".

- 1) Tune the speed controller from the master
 - 1) Turn off the slave axis
 - 2) Autotuning, check parameters with step response
 - 3) Fine-tune parameters manually if necessary
- 2) Tune the speed controller from the slave
 - 1) Turn off the master axis
 - 2) Autotuning, check parameters with step response
 - 3) Fine-tune parameters manually if necessary
- 3) Check parameters with step response from both axes
 - 1) Master: `MTHydPumpController.ControllerMode = mtPUMP_SPEED_CONTROL`
 - 2) Slave: `MTHydPumpController.ControllerMode = mtPUMP_SLAVE_MODE`
 - 3) Check parameters with step response
 - 4) Fine-tune parameters manually if necessary

8.1 Master-Slave exercise

In this exercise, master-slave control is implemented in Automation Studio using Technology Solution HydraulicServoPumpControl.

- Use the previous project with Technology Solution HydraulicServoPumpControl inserted.
- Add another ACOPOS "8V128M.00-2" (slave) in System Designer.
- Add the POWERLINK interface "8AC114.60-2" to slot SS1.
- Connect the "ACOPOS slave drive" with the "ACOPOS master drive" via POWERLINK. The configuration window should then open.
- Configure a resolver "8AC122.60-3" for slot SS2. Slot SS3 can remain unused (skip page).
- Configure motor "8LSA65.EA030D200-1" for motor connection MT1.
- Activate motor simulation.
- Limit switch and quick stop are "normally open"

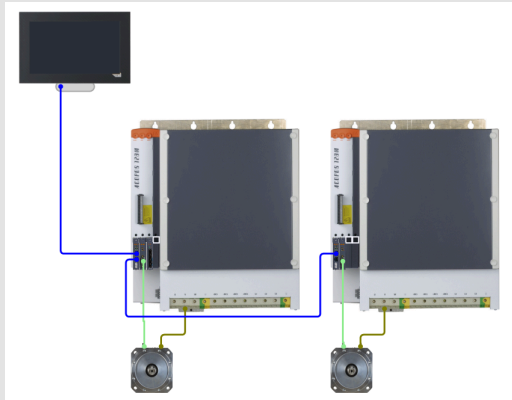


Figure 37: System Designer – Master/Slave hardware concept

- Implement MTHydPumpController in task HydCtrl and configure as slave
 - Axis = Axis reference for the slave axis (gAxis02)
 - MasterAxis = Axis reference for the master axis (gAxis01)
 - ControllerMode = mtPUMP_SLAVE_MODE
 - Configuration.SlaveGearRatio = 1
 - Configuration.MaxPumpAcceleration = 60000.0 rpm/s
 - Configuration.MaxPumpSpeed = 3000.0 rpm
 - Parameters.ControllerParameterSet[0].MaxReversePumpSpeed = -4000.0 rpm
 - Parameters.DisableMinPressureController = 1
- Switch on the master axis using the HMI application and start the controller.
- Switch on the slave axis from the test window
- Switch on MTHydPumpController for the slave axis (Enable = TRUE).
- Slave axis should follow the master axis with ratio SlaveGearRatio.

9 Position control with servo pump drive

Hydraulic servo pump drives can of course also be used for position control. For this to be implemented, a cylinder velocity setpoint must be converted to rotary speed and transferred to MTHydPumpController. Position control is handled by another control block. The following image shows 2 sample concepts for implementing position control using B&R software blocks.

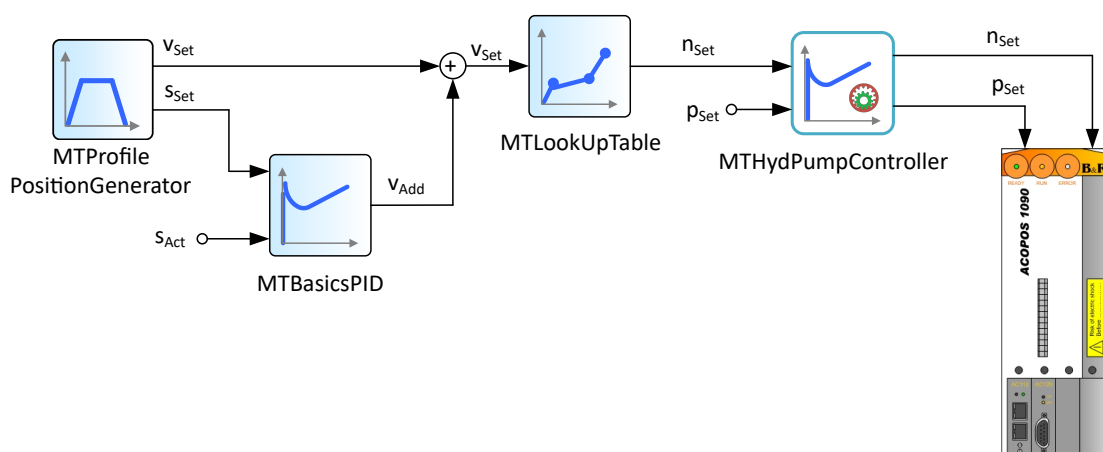


Figure 38: Position control using MTBasicsPID as position controller

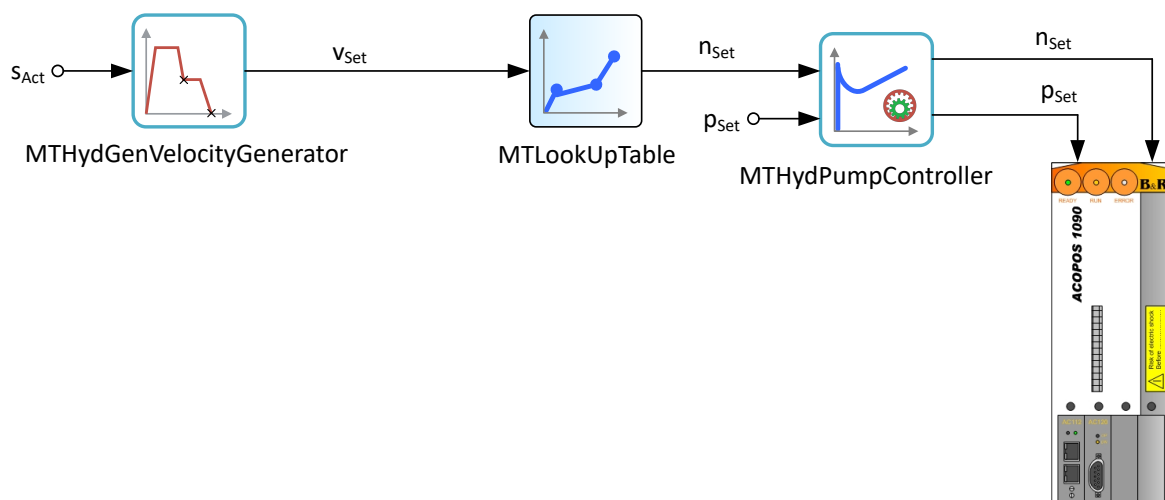


Figure 39: Position control using MTHydGenVelocityGenerator as velocity generator with the function of position-based braking

10 Project development



The following guidelines should be observed. Nevertheless, there are often individual situations and customer requirements that demand a customized solution.

10.1 Project development guidelines

These project development guidelines are meant to serve as a guide for implementation, minimizing potential sources of errors and efficient project management.

- a) Analyze the starting situation.
 - Has the system been sized?
 - Integrated or standalone variants?
- b) Take part in a training and work through this training manual.
- c) Become familiar with the library.
- d) **If necessary, contact the B&R Mechatronic Technologies department.**
- e) Determine customer requirements, collect machine data.
 - Be sure you understand the hydraulic schematics. **If you have difficulties with the hydraulic schematics, please contact the B&R Mechatronic Technologies department or the customer's hydraulic department.**
 - Identify critical/decisive factors related to product quality (e.g. specific cylinder movements, braking processes, etc.).
 - Identify special operating conditions (e.g. oil preheating, setup conditions, long phases where pressure must be maintained).
 - Survey possible control concepts used by the customer.
- f) Document customer requirements in a short but precise specification.
- g) Develop a control concept in the form of a block diagram.
 - Be sure you understand the setpoint profiles for pressure and speed.
 - Identify the active pressure reduction phases.
 - Brakes for closing mechanisms (especially for injection molding).
- h) **If necessary, contact the B&R Mechatronic Technologies department.**
- i) Configure the function blocks
 - What parameters result from the machine data?
 - Collect necessary machine data.
 - Which parameters must be determined during commissioning?
- j) **If necessary, contact the B&R Mechatronic Technologies department.**
- k) Consider the procedures necessary for commissioning the machine.
- l) Consult with the customer.
- m) Implementation
- n) Test on the simulation model.
- o) Installation
- p) Review the project.

10.2 Commissioning instructions

When commissioning a servo pump, there are a few points that should be taken into consideration beforehand in order to eliminate possible sources of error. The following list is intended as a guide during commissioning. A possibility for tuning the control loop is also given.

10.2.1 Commissioning procedure

1. Check the cabling.

- AC131 power supply
- AC131 pressure sensor connection
- POWERLINK / CAN
- Motor cable (encoder, supply)
- Motor fan (24 V or 230 V, direction of rotation at 24 V)
- ACOPOS supply (ground, power mains -> use caution with IT mains)

2. Check the hydraulics.

- Dimensions on the suction side for calculating the maximum acceleration rate (suction hose diameter, suction hose length and pump/oil level difference)
- Calculate or check the acceleration rate MaxPumpAcceleration.
- Set the acceleration rate on the function block.
- Pressure relief valve (must be available)
- Mounting position of the pressure sensor (should be mounted as close to the pump as possible, preferably on the pump itself)
- Oil level
- Pump connections (suction side to the tank, pressure side to the system)
- Discuss the hydraulic schematic with the customer.

3. Direction of rotation of the motor

- Autotune the ACOPOS speed controller.
- When using a resolver, check the filter time (approx. 2 ms).
- Set the speed setpoint SetPumpSpeed to 0 rpm.
- Set the controller MTHydPumpController() to speed control (ControllerMode = mtPUMP_SPEED_CONTROL).
- Start closed-loop servo pump control (previously MCPower, etc.).
- Set the speed setpoint SetPumpSpeed to 10 rpm.
- Check the direction of rotation of the motor using the holes in the coupling housing and compare with the direction arrow on the pump.

If there are no holes, there are two alternative variants:

- a) Close all valves, connect a measurement hose to the pump block and check if oil flows (may take a few minutes).
 - b) Close all valves and check for an increase in the system pressure. The speed can also be increased if necessary since the pressure increases very slowly at low speeds.
- If the direction of rotation is incorrect, it can be inverted using the ReversePumpMotorSpeed parameter. (When using B&R motors with a Dorninger Hytronics DHPH pump, the direction of rotation is inverted in most cases.)

4. Check the pressure relief valve (after the setting has been made).

- Operate the pump at a constant speed (approximately 100–200 rpm).
- Check if the desired system pressure results.
- If necessary, a hydraulic engineer should readjust it.

5. Pump protection

- Cavitation protection

Calculate the MaxPumpAcceleration parameter and set accordingly to limit the acceleration of the pump and the motor. By using the MTHydPumpController function block, the configured axis limits such as v_pos, v_neg, a1_pos and a1_neg are disabled. In Automation Help, there a link to an Excel sheet that can be used to make the calculation.

- Thermal pump protection

The MTHydPumpDriveProtection function block can be used to enable a pump protection feature that estimates the thermal load on the pump in a mathematical model and protects it against overheating. All pumps from Dorninger Hytronics are supported. Pumps from other manufacturers can also be protected as long as their size and displacement volume are known since they will presumably have similar thermal characteristics.

6. Tuning the ACOPOS speed controller

- See section "Tuning the ACOPOS speed controller".

7. Setting the pressure controller

- See section "Setting the pressure controller".

10.2.2 Tuning the ACOPOS speed controller

A prerequisite for functional closed-loop control of pressure and flow rate is a properly tuned speed controller. Important for the speed controller is that it follows the desired speed as precisely as possible and that the output of the speed controller provides a clean setpoint for the flow controller. In order to prevent a permanent control deviation, it is necessary to set an I component for the speed controller. The P component can be determined with the help of ACOPOS autotuning. The I component must be set manually.



Figure 40: Block diagram for tuning the speed controller

Procedure:

- Set the tuning mode to ncSPEED + ncT_FILTER
- Start auto-tuning
- Set integral component of speed controller (default value: Tn = 50 ms).

OPTIONAL: With AS 4.0 and higher, the "Servo Loop Optimizer" can optionally be used for autotuning. This gives you a range of options for compensating resonant frequencies.

Closed-loop speed controller step responses should be recorded in order to check the control loop parameters (if the hydraulic system allows it). All valves must be closed for these tests. The MTHydPumpController function block must be enabled and the minimum pressure controller must be disabled (DisableMinPressCtrl = TRUE). Speed jumps from a standstill (e.g. 0-MaxPumpSpeed) as well as speed jumps under load (e.g. 500-MaxPumpSpeed) should be recorded. These speed jumps are applied to the SetPumpSpeed input of the MTHydPumpController function block.

When tuning the speed controller, all valves are closed and the entire volumetric flow flows directly back into the tank.



Before carrying out the step response tests, the Configuration.MaxPumpAcceleration parameter must be set correctly in order to prevent cavitation of the pump caused by rapid acceleration!



The gain K_v determined by autotuning should be checked for plausibility and adjusted manually if necessary!

The following image shows a block diagram of the ACOPOS control loop. The step response can be recorded with the listed ParIDs using the ACOPOS trace function.

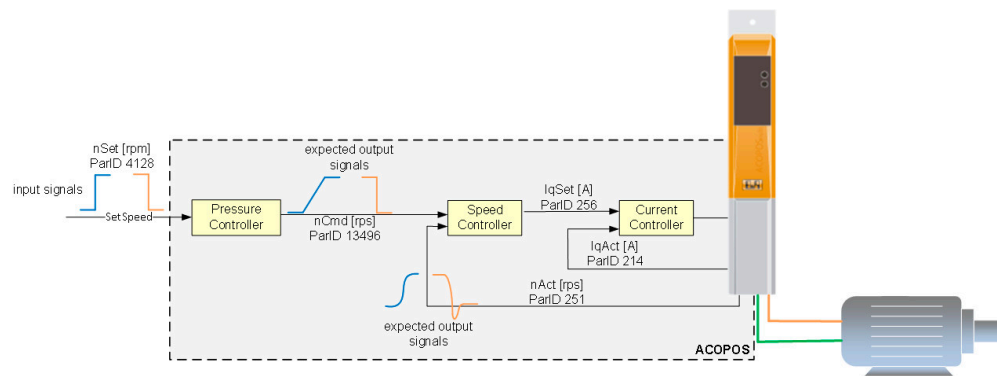


Figure 41: Block diagram for checking the speed controller

10.2.3 Setting the pressure controller

Pressure controller autotuning

The cylinder must always be in the position in which pressure control should be performed, i.e. the cylinder must be pressing against resistance. In addition, the valves must be switched properly so that pressurization can take place accordingly. The servo pump must also be connected to the cylinder chamber responsible for pressurization.

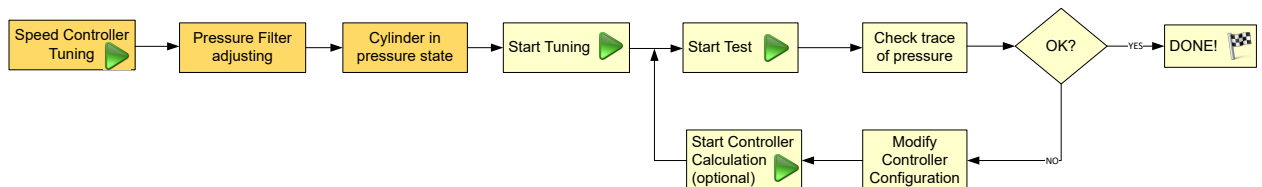


Figure 42: Auto-tuning pressure controller process

Current situation

- Performing speed controller autotuning
- Filter the pressure signal accordingly (e.g. $\text{ActPressureFilterTime} = 0.005 \text{ ms}$).
- Put the cylinder in the position for pressure control.

Tuning

- StartTuning
- TestTuning
- Check pressure signals ActPumpPressure and SetPumpPressure to determine whether dynamic and overshoot requirements are being met. Check manipulated variable ActPumpSpeed to determine whether speed reserves are available.
- If the requirements are not met, the following possibilities exist:

Fine tuning

- Too slow: Reduce RiseTime . This is possible as long as speed reserves exist ($\text{ActPumpSpeed} < \text{Max-PumpSpeed}$).
- Too much overshoot: Change to $\text{ControllerType} = \text{mtPUMP_PID_CONTROLLER}$ and change the strength of the D component with $\text{DerivativeTimeScale}$. If oscillation occurs in the rotary speed, increase FilterTimeScale or reduce $\text{DerivativeTimeScale}$.
- If $\text{FilterRecommended} = \text{TRUE}$, set $\text{ActivateCompensationFilter} = \text{TRUE}$. This should make higher dynamics possible.

Manually configuring the pressure controller

In order to optimally tune the pressure controller, we recommend that the desired axis (cylinder) be moved outside of the movement towards the end position and then maintaining pressure. Pressure jumps can also be carried out. These step responses can be used to tune the control loop so that the behavior meets specific requirements.

The function block must be tuned in pressure control mode ((ControllerMode = mtPUMP_PRESSURE_SPEED_CONTROL). The speed setpoint SetPumpSpeed should be set to an appropriate value (e.g. 1000 rpm). The motor should still be able to accelerate with maximum torque at this speed. If an induction motor is used with field weakening enabled, this is no longer the case. To ensure that no damage occurs during tuning, the pressure limiting valve should be set to a pressure that is non-critical for the system.

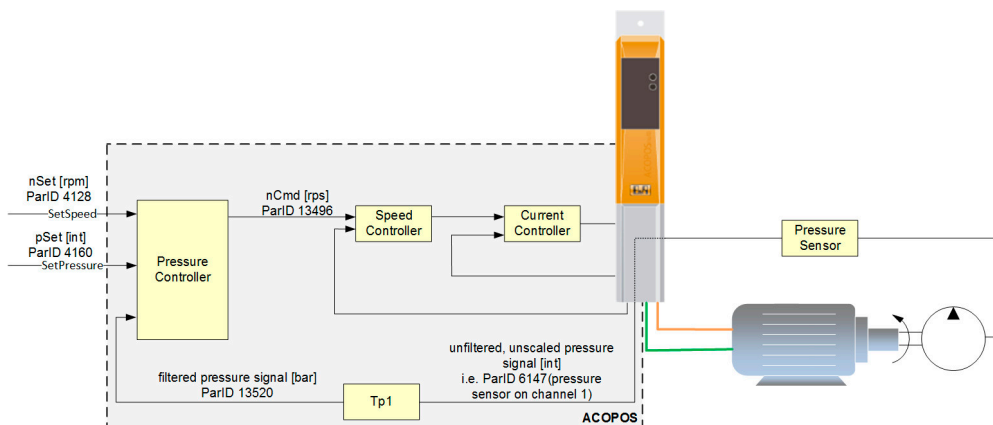


Figure 43: Block diagram for setting the I component



Control loop parameters greatly depend on the hydraulic and mechanical system. The speed controller and pressure controller are configured just once for a machine type. A configuration can include multiple parameter sets.

Procedure

- 1) Set pressure filter (Parameters.ActPressureFilterTime) accordingly: If the filter time is set too low, then signal disturbance will not be sufficiently suppressed (effect: motor doesn't run smoothly). However, setting the filter time too high makes the control sluggish. A good value to start with is Parameters.ActPressureFilterTime = 0.02 s.
- 2) Start with the initial value of the pressure controller's gain factor (K_p), set the desired pressure setpoint SetPumpPressure and let the motor drive the cylinder until it hits the stopper. Then keep increasing K_p until oscillation occurs on the controller output (e.g. louder motor than before) and reduce K_p by 20%. If the initial value causes oscillations, reduce K_p until behavior is stabilized. A good value to start with for K_p is Parameters.ControllerParameterSet[x].PressureController.ProportionalGain = 30 rpm/bar.
- 3) Start with the initial value of the pressure controller's integration time constant (T_i), set the desired pressure setpoint SetPumpPressure and let the motor drive the cylinder until it hits the stopper. Then keep reducing T_i until an undershoot occurs on the measurement value and increase T_i by 10%. If the initial value causes an undershoot, increase T_i until the overshoot disappears from the measurement value. A good value to start with for T_i is Parameters.ControllerParameterSet[x].PressureController.IntegrationTime = 0.1 s.

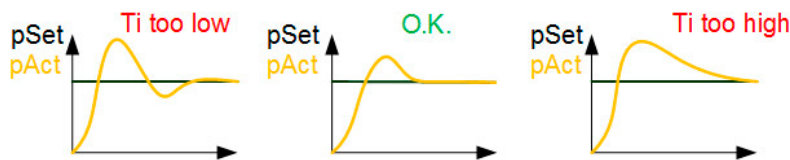


Figure 44: Step responses

- 4) **Optional:** Start with the initial value of the derivative action time constant (T_d) and increase until overshoot is hardly detectable. In addition, each time T_d is increased, the user should also check whether the overshoot is reduced by decreasing T_i . If the overshoot is suppressed too much with the initial value, decrease T_d until the overshoot is hardly detectable. Oscillations that may occur due to increasing T_d can be damped with T_i . A good value to start with for T_d is Parameters.ControllerParameterSet[x].PressureController.DerivativeTime = 0.01 s.
- 5) **Optional:** Overshoot and undershoot can be avoided by configuring a setpoint filter (PT1) for positive (Parameters.ControllerParameterSet.SetPressureFilterTimePosDir) and negative (Parameters.ControllerParameterSet.SetPressureFilterTimeNegDir) setpoint jumps.
- 6) **Optional:** Each cylinder can be tuned separately for each direction of movement as needed. The various parameter sets (SelectControllerParSet) can be used to switch quickly back and forth between the different parameters.

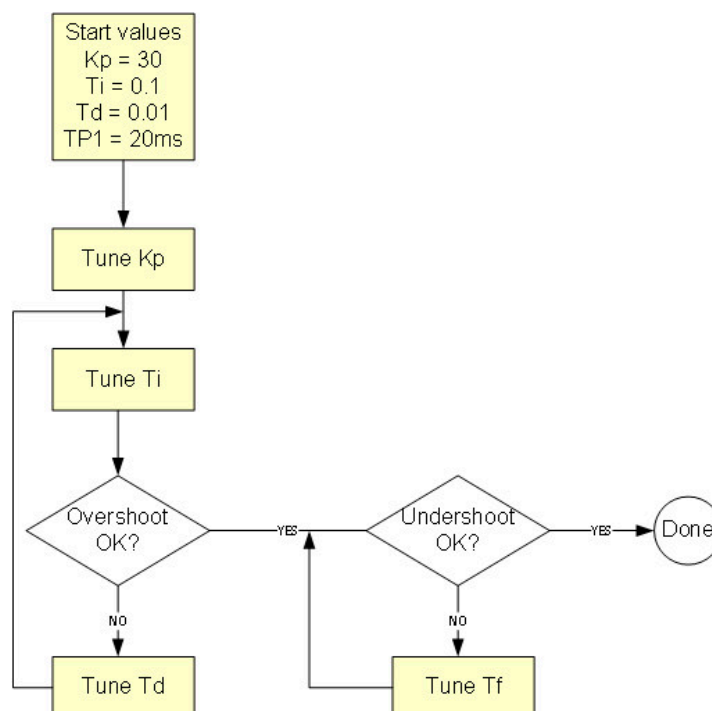


Figure 45: Tuning procedure form the pressure controller

Minimum pressure controller

Good values to start with for the minimum pressure controller parameters are those used for the maximum pressure controller (SetPumpPressure). The same rules 2-4 and 6 used for the maximum pressure controller can be used for tuning.

11 Summary

Combining B&R motion components with efficient internal gear pumps results in an advanced solution to hydraulic tasks that will be effective well into the future. The library and function blocks used for closed-loop servo pump control are easy to use and are an ideal replacement for costly analog controller cards.

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